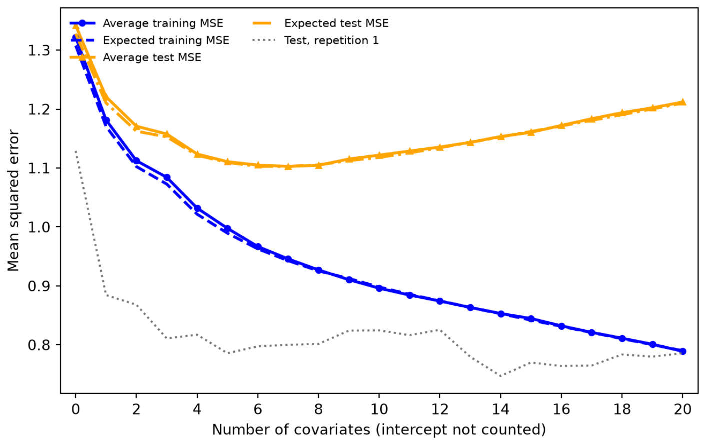
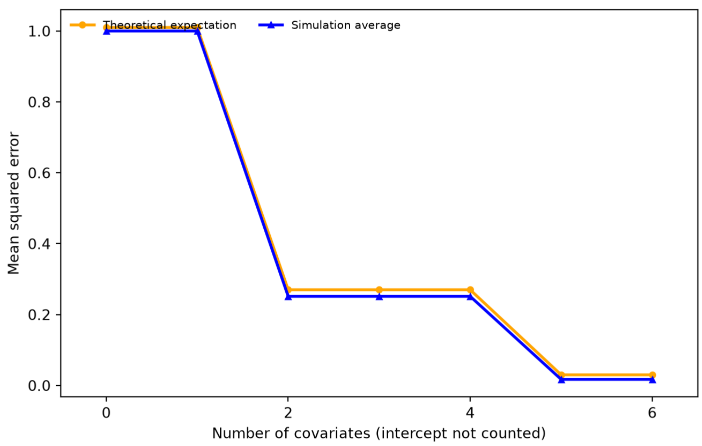
Homework 02

Question 1: Training and test error under a fixed design



Question 2: Prediction error at one target point



When an irrelevant local predictor with $x_0 = 0$ is added to the model, its coefficient-estimation noise does not impact the local prediction because its contribution is multiplied by zero: $b_j x_{0j} = 0$.

Question 3: The optimism correction

Part a

The training residual is

$$\mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu} + (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}.$$

Let $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$. Then $\mathbf{M}$ is symmetric and idempotent. Expanding the squared norm gives

$$\mathrm{MSE}_{\text{train}} = \frac{1}{n}\left(B^2 + 2\boldsymbol{\mu}^\top \mathbf{M}\boldsymbol{\epsilon} + \boldsymbol{\epsilon}^\top \mathbf{M}\boldsymbol{\epsilon}\right).$$

The cross term has expectation zero. Also,

$$E(\boldsymbol{\epsilon}^\top \mathbf{M}\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2\,\mathrm{tr}(\mathbf{M}) = \sigma^2\{n - (p+1)\}.$$

Therefore,

$$E(\mathrm{MSE}_{\mathrm{train}} \mid \mathbf{X}) = \frac{B^2}{n} + \sigma^2\left(1 - \frac{p+1}{n}\right).$$

For the test response,

$$\mathbf{y}^* - \mathbf{H}\mathbf{y} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\mu} + \boldsymbol{\epsilon}^* - \mathbf{H}\boldsymbol{\epsilon}.$$

The test-error noise is independent of the training noise, and the cross terms have expectation zero. Since $\mathrm{tr}(\mathbf{H}) = p+1$
,

$$E\left(\|\boldsymbol{\epsilon}^* - \mathbf{H}\boldsymbol{\epsilon}\|_2^2 \mid \mathbf{X}\right) = n\sigma^2 + (p+1)\sigma^2.$$

Thus,

$$E(\mathrm{MSE}_{\mathrm{test}} \mid \mathbf{X}) = \frac{B^2}{n} + \sigma^2\left(1 + \frac{p+1}{n}\right).$$

Part b

The expected optimism is

$$E(\mathrm{MSE}_{\mathrm{test}} - \mathrm{MSE}_{\mathrm{train}} \mid \mathbf{X}) = \frac{2(p+1)\sigma^2}{n}.$$

For $n = 120$, $p = 4$, $B^2/n = 0.04$, and $\sigma^2 = 1$:

$$E(\mathrm{MSE}_{\mathrm{train}} \mid \mathbf{X}) = 0.04 + 1\left(1 - \frac{5}{120}\right) = 0.9983,$$

$$E(\mathrm{MSE}_{\mathrm{test}} \mid \mathbf{X}) = 0.04 + 1\left(1 + \frac{5}{120}\right) = 1.082,$$

and

$$E(\mathrm{MSE}_{\mathrm{test}} - \mathrm{MSE}_{\mathrm{train}} \mid \mathbf{X}) = \frac{10}{120} = 0.0833.$$

These are expected values over repeated samples. For one realized pair of responses, random variation can cause the test MSE to be smaller than the training MSE.

Part c

$$\widehat{\mathrm{MSE}}_{\mathrm{test}} = \frac{116.4 + 2(4+1)(0.96)}{120} = 1.05.$$

$$C_p = \frac{116.4}{0.96} - 120 + 2(4+1) = 11.25.$$

Their relationship is

$$\widehat{\mathrm{MSE}}_{\mathrm{test}} = \frac{\widehat{\sigma}^2(C_p + n)}{n}.$$

When n and $\widehat{\sigma}^2$ are common to all candidate models, this is an increasing transformation, so minimizing C_p is equivalent to minimizing the corrected test MSE.

Question 4: Comparing Mallows' C_p, AIC, and BIC

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RSS full model: 1089451.7212190419
Sigma^2 hat full model: 3034.6844602201722
RSS full A: 1098908.5656498766
RSS full B: 1089717.0567997107
Cp A: 8.116252959672977
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BIC A: 3005.9482838558592
AIC A: 2974.640259810753
Cp B: 7.087434322792717
BIC B: 3008.754011613133
AIC B: 2973.5324845623886
AIC and BIC fit term: 3.107775248364385
Cp improvement: 3.028818636880295
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Question 5: Validation and final test data

Part a

The misuse begins at step 2 because the final test set is used to compare the candidate models and select m. Reporting the minimum of the eleven test MSE values gives an optimistic estimate of future prediction error because the minimum is likely to benefit from random variation.

Part b

1. Split the training data into five folds.
2. For each candidate predictor count, compute its average MSE across the five validation folds and select the predictor count that minimizes the average five-fold error.
3. Refit the selected model using all the training observations.
4. Use the final test data once to evaluate the selected model.

Part c

The eleven test MSE values and the fact that the smallest one was selected can be reported transparently, but the minimum should not be presented as an untouched final assessment. A new independent batch of test data, or genuinely future observations, is needed for a new final evaluation.

Reference

James, Witten, Hastie, Tibshirani, and Taylor, *An Introduction to Statistical Learning*, Chapters 3, 5, and 6.